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United States: Lehman Brothers Broker-Dealer Emergency Liquidity Program, 2008¹

Ayodeji George² and Steven Kelly^{3,4}

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Abstract

On Sunday, September 14, 2008, a deal to sell the United States investment bank Lehman Brothers Holdings Inc. (LBHI) to United Kingdom-based Barclays fell apart. US authorities informed LBHI that, given the lack of rescue funds, it would need to file for bankruptcy before Monday morning to avoid additional chaos for the firm and markets. However, authorities understood Barclays was still interested in buying Lehman's broker-dealer subsidiary, Lehman Brothers Inc. (LBI). Federal Reserve and Treasury officials were concerned about the impact that the sudden failure of LBI could have on financial markets. LBI had \$87 billion in secured overnight repurchase agreements (repos) that it needed its repo clearing bank, JPMorgan Chase & Co., to unwind Monday morning. For JPMorgan, unwinding those agreements—repaying Lehman's repo lenders and funding the underlying collateral—would have created substantial credit risk during the day without assurance that repo lenders would finance those transactions again that night. The Fed announced on Sunday evening an expansion of eligible collateral at its broad-based Primary Dealer Credit Facility (PDCF); this expansion ensured that LBI could use the facility to allow it to pay back repo lenders after its parent, LBHI, declared bankruptcy. LBI's access to the PDCF was on stricter terms than other primary dealers': It could borrow only on collateral it had already posted with JPMorgan on the previous Friday—and at a bigger haircut than other borrowers. In the early hours of September 15, LBHI filed for bankruptcy, and LBI borrowed \$28 billion through the PDCF. Barclays agreed to take over the Federal Reserve Bank of New York's financing of LBI, in advance of the finalization of its purchase of most of LBI. This replacement transaction took place on September 18, and Barclays borrowed from the PDCF to support LBI. LBI entered Securities Investor Protection Act liquidation proceedings on September 19, and Barclays completed the purchase of most of LBI on September 22, 2008.

Keywords: ad hoc emergency liquidity, Federal Reserve, LBI, Lehman Brothers, primary dealer, Section 13(3)

¹ This case study is part of a Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering ad hoc emergency liquidity provision. A survey of all the cases in this series (Kelly et al. 2025) and the individual cases underlying it are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/vol7/iss1/>. Wiggins et al. (2022) surveys broad-based emergency liquidity programs.

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Overview

This case study is about the ad hoc emergency liquidity assistance the Federal Reserve provided to Lehman Brothers Inc. (LBI), the broker-dealer subsidiary of Lehman Brothers Holdings Inc. (LBHI), during the Global Financial Crisis of 2007–2009 (GFC).

LBHI, commonly known as Lehman Brothers, was the fourth-largest investment bank in the United States, with \$639 billion in assets as of May 2008 (GAO 2011; Wiggins and Metrick 2019). LBHI had thousands of subsidiaries, a structure intended to prevent contagion across the firm from ruptures in any one of its business lines (Wiggins and Metrick 2019). However, stress tests conducted jointly by the Fed and the Securities and Exchange Commission (SEC) in May and June 2008 revealed that LBHI lacked sufficient liquidity or capital to survive an event such as the stress experienced by Bear Stearns earlier that spring. In largely unsuccessful efforts to raise capital following the stress tests, LBHI opened its books to many potential investors, and market participants became increasingly skeptical of its business relative to competitors (Alvarez, Dudley, and Liang 2020).⁵ Uncertainty during the GFC and the announcement of a \$3.9 billion loss in the three months through August 31 caused a sustained run on LBHI, resulting in its bankruptcy filing on September 15, 2008 (Alvarez, Dudley, and Liang 2020; Wiggins and Metrick 2019).

Within the LBHI corporate structure, LBI cleared all US trades and transacted non-US trades of US origin through various LBHI subsidiaries in other countries (Wiggins and

Key Terms

Purpose: To support the tri-party repo market by providing liquidity to LBI through the overnight Primary Dealer Credit Facility, on stricter terms than other borrowers, as a means of facilitating the orderly wind-down of LBI’s repo obligations and the sale of LBI to Barclays	
Announcement Date	Sept. 14, 2008
Operational Date	Sept. 15, 2008
Termination Date	Sept. 18, 2008
Legal Authority	Section 13(3) of the Federal Reserve Act
Administrator	Federal Reserve Bank of New York (FRBNY)
Peak Authorization	Limited to eligible collateral held by LBI as of Friday, Sept. 12 (“Friday criteria”)
Peak Outstanding	\$28 billion
Collateral	Tri-party repo collateral pledged to JPM as of Sept. 12, 2008
Haircut and Recourse	The FRBNY imposed a steeper haircut of 20%, compared to 6.7% for comparable banks
Interest Rate and Fees	2.25%
Term	Overnight
Part of a Package	The FRBNY also provided liquidity assistance using the TSLF and open market operations
Outcomes	LBI entered liquidation on September 19, and Barclays purchased most of LBI’s business on Sept. 22, 2008
Notable Features	Bespoke access terms to an otherwise broad-based facility (the PDCF), to facilitate the orderly wind-down of a broker-dealer

⁵ Lehman raised \$6 billion in capital between June and September 2008. Regardless, the firm announced a \$2.8 billion loss in early June and a \$3.9 billion loss in early September (Alvarez, Dudley, and Liang 2020).

Metrick 2019). LBI primarily financed its operations through repurchase agreements (repos) and was a major player in the overnight tri-party repo⁶ market. Federal Reserve Bank of New York (FRBNY) and Lehman anticipated, correctly, that following LBHI's Chapter 11 bankruptcy⁷ filing, parties that had previously provided LBI with overnight repo financing would refuse to renew the agreements (see Figure 1 for a timeline of the events immediately preceding and following the bankruptcy filing) (Valukas 2010b). The Federal Reserve had established the Primary Dealer Credit Facility (PDCF) in March 2008 to provide overnight secured funding to primary dealers (FRBNY 2008; Fed 2008c). FRBNY originally anticipated that LBI would ultimately require broker-dealer-specific liquidation proceedings but was confident that the wind-down of its repo obligations could be accomplished in an orderly fashion. Barclays also remained an option as a potential buyer for the broker-dealer, despite the breakdown in negotiations for the holding company over the weekend. To support this, the FRBNY committed to provide LBI two weeks of secured overnight financing through the PDCF (Valukas 2010b). The liquidation proceedings outlined by the Securities Investor Protection Act (SIPA) were specific to broker-dealers, where creditors were protected by SIPA insurance up to \$500,000⁸(US Government 1970, 78ccc[2], 78fff-3[a]) . Authorities intended for the PDCF funding to allow LBI to wind down its repo obligations in an orderly fashion, allowing most customers quicker access to their funds than would have been possible under liquidation proceedings (Valukas 2010b). By contrast, an immediate entry into SIPA liquidation might have subjected LBI's repo book to a stay, preventing the wind-down of the portfolio and denying customers access to their funds (Fed 2008a). Then-SEC Chairman Christopher Cox said in a CNN television interview that LBI was

doing business per normal. If you are one of those 83,000 [Lehman customers], you can deal with Lehman Brothers as you always have. We are looking forward to very, very quickly resolving it outside of bankruptcy so that customer cash and customer securities are fully preserved. (Reuters 2008)

On the day that LBHI filed for bankruptcy, September 15, 2008, the Securities Investor Protection Corporation (SIPC) stated that customer accounts at LBI were accounted for, and SIPC's president said that it had not started a liquidation proceeding and did not anticipate doing so (Reuters 2008). The Lehman bankruptcy examiner report said, however:

⁶ A tri-party repo transaction involves a third party, referred to as a clearing bank, facilitating the repo settlement between the borrower and the lender. Clearing banks settle repo transactions on their own books and provide additional aid such as settlement and custodial and collateral management services (Copeland et al. 2012).

⁷ In US bankruptcy code, Chapter 11 bankruptcy provides for the creation of a plan to pay creditors over time before final closure of the firm (USC n.d.).

⁸ SIPA protection covered investors' ownership of securities in the event of broker-dealer liquidation (SIPC n.d.).

The FRBNY anticipated that LBI ultimately would need to be placed in a SIPA proceeding, but that most of LBI's customers would move their accounts to other broker-dealers during the 'orderly unwinding' of LBI, leaving a relatively small portion of LBI's customer accounts when the SIPA proceeding was commenced (which the FRBNY hoped could be delayed until approximately two weeks after the LBHI filing). Without FRBNY funding an orderly unwind of LBI's business, customers might be forced to wait months to gain access to their money in a SIPA proceeding. (Valukas 2010b, 2118)

Reuters reported further: "Under the supervision of the SEC and [Financial Industry Regulatory Authority (FINRA)], Lehman Brothers' customer accounts would be transferred to another brokerage firm. It was unclear how long it would take to transfer the accounts or whether other brokerages will be bidding for them, the regulators said" (Reuters 2008).

After feverish negotiations throughout the weekend of September 12–14, or what would be known as "Lehman weekend," the Fed announced on the evening of Sunday, September 14, 2008, that it was expanding eligible collateral for the PDCF to closely match the assets eligible at tri-party repo clearing banks (Fed 2008c). Though the PDCF was a broad-based FRBNY facility available to all registered primary dealers, the Fed allowed LBI to borrow only on stricter terms. This included what became known as the "Friday criteria" [sic]; this criterion limited the eligible collateral to those assets that LBI had pledged to its tri-party repo clearing bank, JPMorgan Chase & Co., as of Friday, September 12, 2008 (Valukas 2010b). The Fed also applied steeper haircuts owing to the deterioration in LBI's creditworthiness resulting from the bankruptcy of its parent (Baxter 2010b).

Alongside the Chapter 11 bankruptcy of LBHI, the UK's Financial Services Authority (FSA) placed London-based Lehman Brothers International (LBIE) into administration⁹ on September 15 (FSA 2010). This prevented LBI from settling several billion dollars of open trades with LBIE, contributing to LBI's eventual need for liquidation (Wiggins and Metrick 2019).

LBI accessed the PDCF to borrow \$28 billion on September 15, \$19.7 billion on September 16, and \$20.4 billion on September 17 (FRBNY n.d.a). Barclays also lent LBI \$15.8 billion each of those days through the tri-party repo market (Fleming and Sarkar 2014). Discomfort with the FRBNY's exposure to LBI prompted FRBNY officials to request, on September 16, that Barclays take over FRBNY's funding of LBI before the closing of the transaction (Valukas 2010b). Barclays replaced the FRBNY as LBI's primary funding source on September 18, 2008; the same day, Barclays borrowed \$47.9 billion at the PDCF, having not used the facility since May 2008 (FRBNY n.d.a; Valukas 2010b). On September 18, the SIPC announced that it planned to initiate liquidation proceedings for LBI the following day, connecting the decision to a proposed sale to Barclays Capital Inc. (SIPC 2008a). On September 19, 2008, the SIPC placed LBI into liquidation, and Barclays purchased most of LBI's assets early in the proceedings, closing the transaction at a sale hearing on September 22, 2008 (Giddens 2010; Valukas 2010b). LBI was the largest securities brokerage liquidation in US history. SIPC

⁹ The FSA administration process included the appointment of administrators to oversee resolution (PwC 2009).

completed the liquidation in September 2022, having satisfied 111,000 customer claims worth \$106 billion (SIPC 2022).

Figure 1: Timeline of Events Concerning LBI during and after Lehman Weekend

Date	Event
Sep. 9, 2008	LBHI announces \$3.9 billion third-quarter loss.
Sep. 12, 2008 2:50pm	Internal LBHI discussions on the interaction of LBI, LBHI, and Lehman Brothers Special Financing Inc. (an LBI subsidiary) defaults.
Sep. 12, 2008 4pm	LBHI board is informed of management's consultation of bankruptcy counsel, who advises that in case of insolvency, LBI would not qualify for Chapter 11 bankruptcy and would be subject to a SIPA liquidation.
Sep. 13, 2008 11:14am	LBHI is updated that the only remaining potential bidders are Bank of America and Barclays.
Sep. 13, 2008 6:59pm	LBHI prepares a contingency liquidation plan in which LBHI would file for Chapter 11 protection, LBI would enter SIPA liquidation, and LBIE would enter FSA administration.
Sep. 14, 2008 2:15pm	FRBNY official emails colleagues that he and others informed LBHI's bankruptcy counsel that FRBNY had expanded PDCF eligible collateral and that it would be available to LBI unless LBI filed for bankruptcy.
Sep. 14, 2008 7:55pm	LBHI board of directors is informed that the Fed "had expressed its desire" that LBHI file for Chapter 11 bankruptcy as soon as possible in wake of breakdown of Barclays deal (Giddens 2010, Exhibit C, 10).
Sep. 14, 2008 ~8pm	LBI is informed that its PDCF borrowing would be subject to a specific set of rules more stringent than otherwise applied by the facility.
Sep. 14, 2008 9:35pm	<i>Wall Street Journal</i> prints the Federal Reserve Board press release announcing the expansion of PDCF eligible collateral to match assets eligible to be pledged in the tri-party repo market.
Sep. 15, 2008 1:45am	LBHI files for Chapter 11 bankruptcy protection.
Sep. 15, 2008 4:46pm	UK authorities announce that LBIE has been placed into administration.
Sep. 15, 2008 5:16pm	SPIC announces that customer accounts in LBI are safe and that it does not anticipate a SIPA liquidation.
Sep. 15, 2008	LBI borrows \$28 billion from PDCF against \$32.9 billion of collateral in the evening.
Sep. 16, 2008	LBI borrows \$19.7 billion against \$23.3 billion in collateral.
Sep. 16, 2008	LBHI files 8-K report and issues press release announcing agreement to sell LBI to Barclays.
Sep. 16, 2008	The FRBNY requests that Barclays "take out" FRBNY's exposure to LBI before the closing of the transaction (Valukas 2010b, 2161).
Sep. 17, 2008	Barclays agrees to a replacement transaction with FRBNY to become LBI's source of secured funding.
Sep. 17, 2008	LBI borrows \$20.4 billion against \$23.9 billion in collateral from the PDCF.

Sep. 18, 2008	Barclays replaces the FRBNY as LBI's primary funding source; Barclays borrows \$47.9 billion at the PDCF from the Fed against \$50.7 billion of collateral.
Sep. 18, 2008	SIPC issues statement announcing that it will place LBI into SIPA liquidation the following day.
Sep. 19, 2008	SIPC formally places LBI into liquidation.
Sep. 19, 2008	Barclays borrows \$16 billion at the PDCF against \$17.1 billion of collateral.
Sep. 22, 2008	Barclays borrows \$16 billion at the PDCF and completes purchase of LBI.

Source: Authors' analysis.

Summary Evaluation

At the Federal Open Market Committee (FOMC) meeting on Tuesday, September 16, 2008, William Dudley, then head of the Markets Desk at the FRBNY and manager of the System Open Market Account for the FOMC, noted that tri-party repo investors had stayed with Lehman on Friday, September 12, 2008. He attributed this to investors' perception of a lack of rollover risk as long as LBI did not file for bankruptcy over the weekend and to their faith that the Fed, through the PDCF, would be able to serve as the tri-party repo investor of last resort (Fed 2008a). JPMorgan unwound \$87 billion in repo debt on Monday morning, backed by the Fed's expansion of the PDCF. As expected, tri-party repo investors did pull back from LBI during the day on Monday, September 15, after the parent company declared bankruptcy, and the Fed (through the PDCF) and Barclays took their place for the next three days (Fed 2008a; Fleming and Sarkar 2014; Giddens 2010).

First Responders, a book examining US policy responses to the GFC written by the policymakers themselves, discusses the key distinctions that Fed officials made in allowing LBI to access the PDCF. LBI was relatively small, representing less than half of LBHI's assets, and had very few troubled assets. It required a more limited amount of liquidity support and had sufficient valuable assets to serve as collateral (Alvarez, Baxter, and Hoyt 2020). This allowed the Fed to lend to LBI under Section 13(3) of the Federal Reserve Act (FRA), the legal authority on which the Fed had based the creation of the PDCF in March 2008 (Alvarez, Baxter, and Hoyt 2020; Yang 2020).

Laurence Ball of Johns Hopkins University compares the PDCF lending provided to LBI with the counterfactual of whether the program could have been used to lend to LBHI; although LBHI had never been eligible for the PDCF, the "Friday criteria" prevented LBHI from transferring collateral to LBI to post at the PDCF (Ball 2018; Baxter 2010b). He concludes that all of LBHI's immediate borrowing needs could have been offset with PDCF loans. Ball argues that, while Lehman's equity was uncertain, it could have proven more than enough protection for Fed lending when added to LBHI's long-term debt of more than \$100 billion (Ball 2018).

In an interview with the Yale Program on Financial Stability (YPFS), former FRBNY General Counsel Thomas Baxter disputes Ball's claims. The key factor, Baxter says, was that LBHI required an open-ended, unlimited guarantee for any plan that ended with its acquisition by another bank. The Fed could not provide such a guarantee because it would not have been

secured to its satisfaction, a key requirement of its 13(3) authorities. Baxter states that the run on LBHI would have continued regardless of any Fed lending because creditors knew that the company was insolvent. “Plan B,” the PDCF lending to LBI, was a time-limited operation with the goal of keeping its operations going long enough for the wind-down of its trading book (Baxter 2020, 9–10). In Baxter’s words, “One could fault us [the Fed] for maybe not thinking more creatively in August, July, or June to come up with [an] alternative rescue plan, but not in one day on September 14” (Baxter 2020, 10).

Along similar lines, former FRBNY President Timothy Geithner has told YPFS that the Fed was not “lending into a run” with LBI as it would have been had it lent to LBHI, because LBI “was subject to a bankruptcy-like dissolution process pursuant to” SIPA, and LBI’s liquidity needs were more manageable than those of its parent owing to its smaller size and better asset quality; moreover, the Fed was preventing failure until LBI could be acquired by Barclays (Geithner and Metrick 2018, 7–8). He says that most of the assets related to concerns about Lehman’s viability were held by other LBHI subsidiaries (Geithner and Metrick 2018).

More broadly, policymakers—both in real time and in hindsight—have consistently expressed the view that, regardless of LBHI’s potential for collateralizing substantial additional assistance in the manner that LBI did, “lending into a run” in such a way that did not save the firm would not have been an appropriate use of the Fed’s Section 13(3) emergency authority (Baxter 2010a; Bernanke 2015; Geithner 2014; Geithner and Metrick 2018; Lowenstein 2010). As Geithner notes in a 2018 interview:

The legal constraint was that any loan we made had to be reasonably commensurate with the amount of unencumbered collateral the firm could offer. The related, practical consideration was whether such a loan would serve to save the company. With respect to Lehman, we believed it would not. (Geithner and Metrick 2018, 10)

In 2019, he said similarly that “the absence of an explicit solvency test in the Federal Reserve Act does not mean that the lending tools should or could be used to sustain firms that are not viable. To lend freely to the nonviable carries many risks” (Geithner 2019, 22).

As then–Fed Chair Ben S. Bernanke relayed to Lehman’s bankruptcy examiner:

The assessment was that if there was a run, which there would be, the business value would be compromised, and all we would have accomplished would be to make counterparties whole and not succeed in preventing the collapse of the company. (Valukas 2010a, 1504)

Similarly, in response to questions from the Financial Crisis Inquiry Commission (FCIC), then–FRBNY General Counsel Baxter expounded on his view that a bridge loan to LBHI akin to the one given to LBI would have been a “bridge to nowhere,” saying: “This view was not simply my own, but rather at the time was held throughout the US Government, broadly among Lehman’s counterparties, and even by Lehman itself” (Baxter 2010b, 5).

However, the Fed viewed temporarily lending to LBI—as support for the tri-party repo market and to allow for orderly operations until the Barclays deal went through—as consistent with its authority. Baxter added: “Of course, after LBHI filed for bankruptcy, the FRBNY did extend on September 15 an aggregate amount of credit of approximately \$60 billion to LBI, which enabled LBI to continue in business” (Baxter 2010b, 5). As noted, both SIPA and then-SEC Chairman Cox reaffirmed LBI’s going-concern value on its first day of operation following the LBHI bankruptcy (Reuters 2008).

Geithner also disputes Ball’s assertion that LBHI’s long-term debt could have afforded the Fed sufficient protection. In the event that a loan to LBHI did not restore the firm’s viability, Geithner writes that, given policymakers’ then-lack of a resolution regime for nonbanks akin to that for depository institutions,

we could not magically turn that long-term debt into equity to absorb the losses. The rest of the liabilities might bleed away, some quickly, some slowly . . . If the Fed had assumed all the risk in lending to Lehman, and we had to replace the holders of \$500 billion of [non-long-term] runnable debt, would we have been able to capture the remaining value in Lehman’s assets and businesses ahead of the other claimants to those assets, and would that value be sufficient to cover our exposure? In the extreme case, the Fed would be the holder of assets previously valued at \$600 billion but now worth substantially less, having paid out the \$500 billion of Lehman’s other obligations and with the core business of the bank deeply damaged. (Geithner 2019, 23)

Context: United States 2008–2009	
GDP (SAAR, nominal GDP in USD)	\$14.8 trillion in 2008
	\$14.5 trillion in 2009
GDP per capita (SAAR, nominal GDP in USD)	\$48,570.1 in 2008
	\$47,195.0 in 2009
Sovereign credit rating (long-term debt)	Moody's: Aaa in 2008
	S&P: AAA in 2008
	Fitch: AAA in 2008
	Moody's: Aaa in 2009
	S&P: AAA in 2009
	Fitch: AAA in 2009
Size of banking system	\$10.1 trillion in 2008
	\$9.1 trillion in 2009
Size of banking system as a % of GDP	68.4% in 2008
	62.7% in 2009
Size of banking system as a % of financial system	31.9% in 2008
	28.5% in 2009
Five-bank concentration of banking system	44.9% in 2008
	44.3% in 2009
Foreign involvement in banking system	18% in 2008
	19% in 2009
Existence of deposit insurance	Yes in 2008
	Yes in 2009

Sources: Bloomberg; World Bank Deposit Insurance Dataset; World Bank Global Financial Development Database.

Key Design Decisions

- 1. Purpose: The Fed allowed LBI to access PDCF—on stricter terms than it offered other borrowers—to support the tri-party repo market and facilitate an orderly wind-down and sale of LBI’s operations.**

The Fed provided expanded PDCF financing to LBI because “the FRBNY and Lehman believed, correctly, that . . . parties that previously had provided Lehman with overnight secured financing through repos would refuse to ‘roll’ (i.e., renew) those repos beginning on Monday, September 15” (Valukas 2010b, 2118–19).

The FRBNY established the Primary Dealer Credit Facility on March 16, 2008, to provide overnight secured financing to primary dealers.¹⁰ The goal of the program was to support the orderly functioning of repo markets and financial markets more generally in the early stages of the Global Financial Crisis (FRBNY 2008). For a full account of the PDCF as a broad-based program, see Yang (2020).

Lehman Brothers Inc. was the broker-dealer subsidiary of Lehman Brothers Holdings Inc. (Wiggins and Metrick 2019). LBHI suffered a sustained lack of confidence in the wake of the Bear Stearns rescue owing to the similarity of its business model to the rescued investment bank (Alvarez, Dudley, and Liang 2020). Starting in the summer and continuing through the fall of 2008, Fed officials considered the ramifications of a possible Lehman bankruptcy. According to the Financial Crisis Inquiry Report (the FCIC report), which was written to present the conclusions of the Financial Crisis Inquiry Committee, the fundamental issue was that the SEC would want LBI to continue functioning after a bankruptcy (FCIC 2011). Given the legal constraint that any loan be reasonably commensurate with the amount of unencumbered collateral the firm could offer, and a view among policymakers that a loan should be made only in the event it saved the firm, a bridge loan to LBI was viewed as viable, whereas a loan to LBHI was not (Geithner and Metrick 2018). Further, authorities considered the bridge loan to LBI to be a key measure to contain the damage of an LBHI bankruptcy (Bernanke 2015; Geithner 2014; Paulson 2010). A loan through the PDCF would allow the broker-dealer to work down its trading book, facilitating a more orderly wind-down, inclusive of a sale of much of the business to Barclays, before the broker-dealer’s liquidation through the SIPA process (Bernanke 2015; Geithner and Metrick 2018). SIPA liquidations for US broker-dealers are managed by the SIPC, a government-created nonprofit funded by the investment industry (SIPC 2009).

The goal was to delay the SIPA liquidation for a few days to allow LBI to pay back its repo lenders and allow Barclays to purchase most of the broker-dealer (Bernanke 2015). If the firm had immediately entered SIPA liquidation on September 15, Lehman’s repo obligations could have been subject to a stay as part of the proceedings. In an automatic stay, a failed company’s counterparties would be unable to seize collateral for some period of time. Repo lenders were exempt from the automatic stay provisions of the US bankruptcy code, but they

¹⁰ The FRBNY defines primary dealers as its counterparties in the implementation of monetary policy (FRBNY n.d.b)

would not necessarily be exempt from the stay in a SIPA liquidation given the rights held by the SIPC (Bane et al. 2008).¹¹ Indeed, when LBI was liquidated later in the week, SIPA imposed a 21-day stay on its remaining repo lenders (Giddens 2010).

On the weekend of September 12–14, Lehman weekend, authorities faced the reality that LBHI would need to file for bankruptcy after an unsuccessful sale of LBHI to Barclays (FCIC 2011). Whatever the condition of the parent company, then-Fed Chairman Bernanke and other officials at the Fed and SEC considered LBI, the broker-dealer, to be solvent, and authorities were confident that the broker-dealer could achieve an orderly wind-down of its operations in the wake of the holding company's bankruptcy filing (Bernanke 2015). Moreover, officials thought a Barclays deal would still happen for the bulk of LBI. However, JPMorgan, LBI's clearing bank, had informed authorities earlier in the weekend that it would not be willing to unwind LBI's repo positions unless the Fed expanded the types of collateral that could be financed through the PDCF to include all types of collateral that JPMorgan and Bank of New York Mellon (BNY Mellon), the other tri-party repo clearing bank, accepted. In tri-party repo transactions, the tri-party repo clearing banks took a substantial amount of intraday risk. Each morning, clearing banks returned cash to lenders and took possession of borrowers' collateral, known as "unwinding" repo loans (FCIC 2011, 284). Thus, the clearing banks took significant credit risk on those positions until the evening, when they matched lenders and borrowers for the next day's loans. That weekend, JPMorgan was worried about continuing to provide intraday credit in the event repo lenders pulled back from Lehman, if the Fed was not going to make PDCF funding available on all of the collateral it held for Lehman (FCIC 2011).

On September 14, 2008, the Fed did expand the types of eligible collateral for the PDCF to closely match collateral accepted by tri-party repo clearing banks (Fed 2008c). Having learned of this, JPMorgan officials communicated internally that it would continue to provide clearing and settlement services as long as LBI "was doing an orderly liquidation" (Valukas 2010b, 2121). JPMorgan would continue to unwind the tri-party book given that LBI was not filing, and it would post any end-of-day LBI positions to the PDCF (Buyers-Russo 2008a; Buyers-Russo 2008b). JPMorgan did unwind \$87 billion in trades for Lehman on the morning of Monday, September 15, and held those positions until the evening. However, JPMorgan's position on lending to Lehman became more unclear on Monday, prompting continued correspondence between FRBNY officials and JPMorgan (Valukas 2010b). By mid-morning Monday, JPMorgan clarified that it would continue to serve as LBI's clearing bank but would limit its aggregate payments settlement exposure to LBI to \$1 billion (Buyers-Russo 2008a; Buyers-Russo 2008b; Valukas 2010b).

The PDCF was a broad-based FRBNY facility available to all registered primary dealers. However, the FRBNY limited LBI's eligible collateral to those assets which had been pledged to its tri-party repo clearing bank, JPMorgan, as of September 12, 2008—referred to as the "Friday criteria" (Valukas 2010b, 2119–20). According to then-FRBNY General Counsel

¹¹ In a SIPA liquidation, the SIPC held the right "to seek a stay of foreclosures on or other dispositions of securities collateral, securities sold by a debtor under a repurchase agreement or securities loaned under a securities lending agreement" (Bane et al. 2008).

Baxter, the Friday criteria satisfied the FRBNY's need to verify that none of the collateral pledged to the PDCF could be subject to a superior claim in LBHI's bankruptcy proceedings. He added that the criterion also ensured that PDCF collateral would not be subject to claims of fraudulent conveyance.¹² In protecting against both scenarios, the Fed intended for the Friday criteria to prevent the FRBNY from being exposed to any situation in which it may have been under-secured (see Key Design Decision No. 10, Balance Sheet Protection). In addition to the Friday criteria, the FRBNY assigned steeper haircuts to LBI's assets owing to the bankrupt status of its parent (Baxter 2010b).

The FRBNY committed to provide LBI with access to the PDCF to facilitate an orderly wind-down of its operations in advance of a probable sale of most of LBI to Barclays and SIPA liquidation proceedings thereafter. FRBNY authorities anticipated that most customers would be able to transfer their accounts to other broker-dealers during an orderly unwinding period. This would give customers more immediate access to their money than the several months it would take to gain access in a SIPA proceeding (Valukas 2010b). LBI borrowed from the PDCF daily from September 15 through September 17 (FRBNY n.d.a).

Lehman Brothers International contacted the UK Financial Services Authority on the evening of September 14, 2008, advising that administration was necessary. The FSA then placed LBIE into administration on September 15 (FSA 2010). This prevented LBI from settling billions of dollars in open trades with its international subsidiary, making its orderly wind-down more difficult (Wiggins and Metrick 2019). The Fed said that it was not possible to provide credit access to LBIE, though it later extended credit to the UK broker-dealer subsidiaries of Goldman Sachs, Morgan Stanley, and Merrill Lynch, starting on September 21 (Fed 2009c; FSA 2010). When extending that credit, authorities cited the unusual and exigent circumstances and the need for support during the transition to managing their funding within the bank holding company structure (Fed 2009c).

Barclays agreed to acquire most of LBI's operations on September 16, 2008 (LBHI 2008). The FRBNY then reached an agreement on September 17 in which Barclays became LBI's source of secured funding to reduce the US government's exposure (Valukas 2010b). Barclays borrowed \$47.9 billion from the PDCF on September 18 and \$16 billion on each of September 19 and 22 (all on an overnight basis) to support LBI until the acquisition deal closed on September 22, 2008; its daily PDCF borrowings gradually declined to \$8 billion on September 30 and zero by the end of October (Fleming and Sarkar 2014; FRBNY n.d.a).

In their memoirs of the crisis, former Fed Chair Bernanke, FRBNY President Geithner, and Treasury Secretary Henry M. Paulson, Jr., all cite the preservation of the tri-party repo market as the key rationale for using the PDCF to lend to LBI (Bernanke 2015; Geithner 2014; Paulson 2010). The Fed was able to lend to LBI because its liquidity needs and size were much smaller than those of its parent's, and the value of its collateral was sufficient to cover its immediate funding needs and allow its business to continue functioning as it unwound its positions (Geithner and Metrick 2018). In his book recounting the events, Bernanke calls LBI

¹² Fraudulent conveyance refers to the transfer of property to prevent or delay the payment of debt to a creditor without the creditor's knowledge (Cornell n.d.)

“relatively healthy,” and adds that the PDCF loans helped facilitate LBI’s sale to Barclays, which would in turn repay the Fed upon completion of the acquisition (Bernanke 2015, 268–69).

The minutes of the FOMC emphasize the focus in the moment on preserving the tri-party repo market. Then–manager of the System Open Market Account for the FOMC Bill Dudley said that the Federal Reserve could serve as the tri-party repo lender of last resort as long as LBI did not file for bankruptcy during Lehman weekend (Fed 2008a). The following quote makes this point clear:

Lehman’s experience suggests to me that, if we can contain the broad parameters of this crisis so that it doesn’t spread much further, then we can keep the tri-party repo investors from bolting because they don’t really have a huge amount of risk as long as we are there behind them to take them out when their overnight obligation comes due the next day. (Fed 2008a, 8–9)

2. Part of a Package: LBI received liquidity support from multiple Fed operations to facilitate its eventual sale to Barclays.

Alongside PDCF borrowing, LBI also received liquidity from the Term Securities Lending Facility (TSLF) and the Fed’s open market operations (OMOs) (Valukas 2010b).¹³ On September 15, 2008, the FRBNY provided \$18.5 billion to LBI using the TSLF¹⁴ and \$2.8 billion using OMO overnight trades that occurred every night (Alvarez & Marsal 2008). LBI continued to access these facilities until September 18. Barclays also lent LBI \$15.8 billion overnight from September 15 to 17 (Fleming and Sarkar 2014). On September 18, Barclays replaced the Fed as LBI’s source of secured funding (Valukas 2010b).

On September 18, 2008, Barclays Capital Inc. began to provide the necessary funding for LBI to continue to wind down its tri-party repo operations (see Key Design Decision No. 4, Administration). Figure 2 shows the borrowing from the PDCF and TSLF by Barclays in the days following the agreement, up to the formal purchase of LBI on September 22, 2008.

¹³ The Federal Reserve had started using temporary OMOs in the second half of 2007 as a means of providing the banking system with additional reserves while keeping the federal funds rate near its target (English and Mosser 2020). By fall 2008, OMOs and other conventional tools were already heavily engaged, contributing to the Fed’s use of its Section 13(3) lending authority (English and Mosser 2020).

¹⁴ A former FRBNY official involved in the LBI assistance told YPFS that, while the Fed charged LBI higher haircuts on PDCF loans, it did not do so for the TSLF owing to the higher-quality collateral required to borrow from the TSLF.

Figure 2: PDCF and TSLF Borrowing by Barclays, September 18–22, 2008

Loan Date	Source of Funding	Type of Funding	Amount (Billions of Dollars)	Share of Collateral Pledged (Percent)							
				UST/ Agency Securities	Agency MBS	Private-Label MBS	Corporate Bonds	Municipal Bonds	ABS	Equity	Other ^a
09/18/2008	Fed	PDCF	47.9	14.7	52.8	5.9	7.4	0.5	3.1	15.5	0.1
09/18/2008	Fed	TSLF	5.0	0.0	45.4	20.1	0.0	0.0	34.4	NE	NE
		Schedule 2									
09/19/2008 ^b	Fed	PDCF	16.0	0.8	10.3	11.9	20.3	2.7	3.4	50.4	0.3
09/19/2008 ^c	Fed	TSLF	2.7	0.0	2.0	35.6	27.4	0.0	34.4	NE	NE
		Schedule 1									
09/22/2008 ^d	Fed	PDCF	16.0	0.4	11.0	10.4	20.4	2.7	3.7	51.1	0.3

Source: Fleming and Sarkar 2014.

3. Legal Authority: Section 13(3) of the Federal Reserve Act authorized the PDCF, and the requirement for “satisfactory security” resulted in the specific collateral terms for LBI.

The Fed used the authority granted by Section 13(3) of the Federal Reserve Act to establish the PDCF (Yang 2020). This authority was premised on “unusual and exigent circumstances” and required the Fed to be “secured to the satisfaction” of Fed officials (US Government 2008). Federal Reserve officials have defined this stipulation to mandate that Fed authorities are reasonably certain *ex ante* that the central bank will be repaid in full. The authority is also considered to be discretionary, allowing interpretive space to determine the level of security necessary to be “satisfied” (Alvarez 2022, 5–6).

Former Federal Reserve Board General Counsel Scott Alvarez emphasizes the discretionary nature of 13(3) lending; on the decision not to lend to LBHI, he says, “we just didn’t feel we could take the losses that would be associated with that, given the authority that we had” (Alvarez 2022, 5, 19). Former FRBNY General Counsel Thomas Baxter cites the necessity of satisfactory security as the reason that authorities did not provide an unsecured and unlimited guarantee to LBHI. He asserts that creditors knew that LBHI was insolvent, and that the Fed would have been lending into an ongoing run from a bank that could not be saved (Baxter 2020).

As discussed in the Overview, the view of the inability for the loan to save all of LBHI proved important in policymakers’ determination of appropriate use of their Section 13(3) authority. Baxter testified to the FCIC that [emphasis added], “In this case, Lehman had no ability to pledge the amount of collateral required to **satisfactorily secure** a Fed guarantee, one **large enough to credibly withstand a run** by Lehman’s creditors and counterparties” (Baxter 2010a, 9). Former FRBNY President Timothy Geithner later writes, “The Fed’s emergency authorities limited how much risk we could take; we were the central bank of the United States, and we weren’t going to defy our own governing law to lend into a run” (Geithner 2014, 187). Geithner also has since pointed out that “the absence of an explicit solvency test in the Federal Reserve Act does not mean that the lending tools should or could be used to sustain firms that are not viable. To lend freely to the nonviable carries many risks” (Geithner 2019, 22).

LBHI's outcome thus contrasts with the Fed's ad hoc assistance in the rescue of Bear Stearns, which, unlike LBHI in the Fed's view, had going-concern value, as demonstrated by the fact that it was being acquired by JPMorgan. In that case—of having a viable acquirer—Geithner says that the Fed's loan met its “secured to the satisfaction” standard because “if we held on to the assets for a few years, we would probably break even”—that is, the Fed would have an ex ante expectation of full repayment (Geithner 2014, 156; US Government 2008). Similarly, LBI appeared to have a likely, though not immediate, buyer in Barclays (Valukas 2010b).

Geithner says that authorities were reasonably confident in lending to LBI largely owing to its smaller size and the quality of collateral—as well as its SIPA resolution independent of the bankruptcy process. LBI's liquidity needs were much smaller than LBHI's, and the value of its collateral was sufficient to cover its immediate financing needs and allow its business to continue functioning as it unwound its positions. He says that most of the assets that caused concern over LBHI's viability were held outside LBI (Geithner and Metrick 2018). Additionally, the Friday criteria served as a key legal mitigant, preventing a situation in which the FRBNY could have been under-secured (see Key Design Decision No. 10, Balance Sheet Protection) (Baxter 2010b).

As Geithner writes in a 2019 article:

We were able to act to help prevent a precipitous liquidation of the broker-dealer because Lehman's U.S. broker-dealer affiliate was much smaller than the firm as a whole, its liquidity needs were smaller, and we concluded that the value of the collateral held in that entity was sufficient to cover its immediate funding needs. The funding ... meant that this smaller entity could continue to close out its trades for a few more days, shrinking the firm and moving towards a more orderly liquidation. (Geithner 2019, 22)

By contrast for the holding company, “without a willing buyer and without access to a resolution regime like the FDIC's regime for banks [see the Summary Evaluation], we believed that to lend on a scale necessary to save it would be outside the limits of what we could do” (Geithner 2019, 23).

4. Administration: The FRBNY administered the PDCF, which functioned as a backstop to the tri-party repo market.

The FRBNY administered the PDCF, providing secured funding to LBI between September 15 and 17 (FRBNY 2009; FRBNY n.d.a). JPMorgan priced LBI's TSLF and PDCF collateral, but not that which was used for OMO (see Key Design Decision No. 2, Part of a Package) (Alvarez & Marsal 2008).

Despite the Fed's willingness to provide funding, some FRBNY officials were concerned about its exposure to LBI given the two-week funding time frame. On September 16, the FRBNY contacted Barclays and stated that for LBI to continue accessing PDCF funding, Barclays needed to remove FRBNY's exposure to LBI before the closing of the sale agreement (Valukas 2010b). Barclays's US based broker-dealer, Barclays Capital Inc., already had access to the PDCF as a primary dealer (FRBNY n.d.a; GAO 2011).

On September 17, 2008, Barclays and the FRBNY came to an agreement (the replacement transaction), which stated that Barclays would take over as LBI's source of secured funding¹⁵ (LaRocca 2008). In exchange, Barclays would receive the LBI securities that had been pledged to the FRBNY, including residential mortgage-backed securities that were not part of the asset purchase agreement being negotiated between Barclays Capital Inc. and LBI. The replacement transaction unwound FRBNY's financing of LBI on September 18, with FRBNY delivering LBI collateral to LBI's clearing box¹⁶ at JPMorgan in exchange for a \$46.2 billion advance. LBI and Barclays then entered into a tri-party repo transaction through JPMorgan (Valukas 2010b). During the period between the replacement transaction and the closure of the sale (September 18–22), Barclays borrowed \$47.9 billion on September 18, \$16 billion on September 19, and \$16 billion on September 22 (FRBNY n.d.a; Fleming and Sarkar 2014).

5. Governance: The PDCF was subject to all oversight procedures outlined in Section 13(3).

The PDCF was subject to the oversight rules outlined in the FRA, which stipulated only that emergency lending facilities were subject to “limitations, restrictions, and regulations” as the Fed's Board of Governors may define (US Government 2008). The Government Accountability Office (GAO) mentions the PDCF assistance to Lehman in its review of the Fed's emergency lending activities during the GFC. The report emphasizes that though LBHI's CEO served on the board of the FRBNY, the firm did not receive individual assistance to prevent its failure (GAO 2011).

The Fed mentioned the PDCF in its February 2009 Monetary Policy Report to the Congress, although it did not mention the LBI-specific terms (Fed 2009a). The Committee on Banking, Housing, and Urban Affairs of the Senate (Senate Banking Committee) and the Committee on Financial Services of the House of Representatives (House Finance Committee) exercised oversight over the Federal Reserve (US Government 2008).

The trustee overseeing LBI's SIPA liquidation regularly met with and provided information to US regulatory bodies, including the FRBNY, SEC, FINRA,¹⁷ and the Commodity Futures Trading Commission (CFTC). The SIPC considered coordination with government to be very important, and the trustee received and responded to requests for information from dozens of federal, state, and local government agencies (Giddens 2010).

Section 125 of the Emergency Economic Stabilization Act (EESA) passed on October 3, 2008, established the Congressional Oversight Panel and tasked it with reviewing the state of financial markets and regulation (COP 2011; US Congress 2008, sec. 125). Section 129 of the EESA required the Fed to report to the Senate Banking Committee and the House Finance Committee within seven days of any use of Section 13(3) lending authority (US Congress

¹⁵ Barclays had already been providing LBI with liquidity through tri-party repo (see Figure 3).

¹⁶ The term “clearing box” refers to the primary dealer's allocation of securities used in tri-party repo operations (Copeland et al. 2012).

¹⁷ FINRA is a private, not-for-profit organization that oversees US broker-dealers with the goal of promoting market confidence (FINRA n.d.).

2008, sec. 129). The law also required that the Fed provide updates every 60 days while the 13(3) loan was outstanding (US Congress 2008, sec. 129). The Fed compliantly issued periodic reports on its outstanding lending facilities, the first of which was released on December 29, 2008 (Fed 2008d). Its February 2009 report on outstanding lending facilities mentioned the broad-based usage of the PDCF but did not mention LBI specifically (Fed 2009b).

6. Communication and Disclosure: Authorities communicated that LBI's operations would continue, but the question of whether LBI would require SIPA liquidation was initially unclear.

On September 14, 2008, the Fed announced revisions to the PDCF, expanding its criteria to closely match collateral accepted in the tri-party repo market (Fed 2008c).

On September 15, 2008, the day that LBHI filed for bankruptcy, the Securities Investor Protection Corporation stated that customer accounts at LBI were safe (Reuters 2008). The SIPC policy stated that brokerage firms were required to keep customer cash and securities separate from company accounts. The SIPC president stated that LBI had done this properly and said the SIPC would not take immediate action but play a standby role (Pitt 2008). Because all customer property was accounted for, the SIPC stated that it had not started a liquidation proceeding and did not anticipate doing so. FINRA also sought to reassure the public, stating that LBI was in good standing (Reuters 2008). CME Group Inc., a large derivatives exchange, said that LBI continued to meet all its obligations to CME Clearing (AP 2008). The SEC also released a statement emphasizing that LBI had not been part of LBHI's bankruptcy, that customer assets were safe, and that SEC staff would "oversee the orderly transfer of customer assets to one or more SIPC-insured brokerage firms" (SEC 2008).

Lehman Brothers issued a press release and LBHI filed an 8-K report on September 16, 2008, announcing that Barclays had agreed to acquire most of LBI's business (Lehman Brothers 2008; LBHI 2008). The press release also stated that Barclays would provide a "substantial interim credit facility" to fund LBI's daily operations (Lehman Brothers 2008, PDF 2).

On September 18, 2008, the SIPC issued a statement saying that it would place LBI into liquidation the following day, September 19. The statement said that authorities made the action in connection with the proposed sale to Barclays, which was also scheduled for a hearing on September 19 (SIPC 2008a). On September 20, the SIPC communicated that under the asset purchase agreement negotiated in Bankruptcy Court, Barclays was to acquire many of LBI's business assets. Customer accounts were transferred to Barclays Capital Inc. or a separate trading platform (SIPC 2008b). Customer claims left in LBI for SIPA liquidation proceedings originally amounted to \$23 billion (SIPC 2022).

7. Source and Size of Funding: The FRBNY used the PDCF to lend \$28 billion to LBI across three days.

The FRBNY committed to support the daily financing needs of LBI for up to two weeks (Valukas 2010b). The size of the loans was limited to the amount of eligible collateral that satisfied the Friday criteria (see Key Design Decision No. 10, Balance Sheet Protection)

(Baxter 2010b; FRBNY 2009). On September 15, 2008, LBI borrowed \$28 billion, followed by borrowing of approximately \$20 billion on September 16 and 17 (FRBNY n.d.a). LBI had \$73 billion in liabilities on its estimated balance sheet on September 15, 2008 (Valukas 2010b).

On September 17, 2008, Barclays agreed that it would fund LBI's daily operations until the sale closed (LaRocca 2008; Valukas 2010b). Barclays Capital Inc. borrowed \$47.9 billion from the PDCF on September 18 and \$16 billion on September 19 and 22 (FRBNY n.d.a; Fleming and Sarkar 2014).

Former FRBNY President Geithner says that the relatively small size of the lending was one of the considerations when choosing to lend to LBI and not LBHI (Geithner 2019; Geithner and Metrick 2018). Former FRBNY General Counsel Baxter and Fed Board General Counsel Alvarez echo that view, writing with a coauthor, "Importantly, Lehman's broker-dealer needed a more limited amount of financing" (Alvarez, Baxter, and Hoyt 2020, 159).

8. Rates and Fees: The PDCF charged the FRBNY's primary credit rate.

The PDCF charged the FRBNY's primary credit rate, which was 2.25% (FRBNY n.d.a; FRBNY 2009).¹⁸ At the time, "Regulation A"—the rules written by the Fed to govern its provision of liquidity under Section 13(3), among other things—called for all Section 13(3) lending to be provided at an interest rate "above the highest rate in effect for advances to depository institutions" (Fed 2008b, 201.4[d]). This would suggest a rate higher than the primary credit rate (indeed, perhaps higher than the secondary credit rate available to banks at a premium to the primary credit rate) (Kelly 2021). However, the Fed regularly made exceptions to this rule in providing emergency credit during the GFC (Fed 2009c; Kelly 2021).

9. Loan Duration: The FRBNY agreed to provide up to two weeks of overnight secured financing.

During Lehman weekend, the FRBNY advised LBI that it would provide up to two weeks¹⁹ of overnight secured financing through the PDCF to support an orderly wind-down of LBI's repo funding (Valukas 2010b). LBI accessed the PDCF between September 15 and 17 (FRBNY n.d.a). The FRBNY's replacement transaction with Barclays unwound FRBNY's financing of LBI on September 18, 2008 (LaRocca 2008; Valukas 2010b).

10. Balance Sheet Protection: The PDCF charged LBI a higher haircut than it charged other borrowers and limited the tri-party-repo-eligible securities that LBI could pledge under the Friday criteria.

On September 14, 2008, the Fed announced an expansion of collateral eligible to be pledged at the PDCF. The announcement stated that eligibility had been broadened to "closely match"

¹⁸ PDCF borrowers who accessed the PDCF on more than 30 out of 120 business days were also charged a frequency-based fee (FRBNY 2009).

¹⁹ As previously mentioned, some at the FRBNY believed that it was risking too much exposure with the two-week funding time frame (Valukas 2010b).

collateral eligible to be pledged in the tri-party repo systems of the two major clearing banks (Fed 2008c). The PDCF provided loans with recourse to the recipient's other assets (Yang 2020).

The FRBNY, in what became known as the Friday criteria, did not allow for intra-Lehman collateral transfers to increase LBI's borrowing capacity at the PDCF. The criterion obligated LBI to certify that it owned the securities that it pledged to the PDCF as of September 12, 2008, that the securities had not been transferred by LBHI. Authorities considered this certification to be an important mitigant of legal risk given LBI's situation as a subsidiary of a bankrupt parent. Then-FRBNY General Counsel Thomas Baxter said that if LBHI had been able to transfer securities to LBI to be pledged at the PDCF, those securities could have become subject to preference or conveyance claims by creditors. This would have left the FRBNY under-secured. A series of emails involving FRBNY officials and Lehman employees confirmed that LBI's PDCF collateral matched the type of assets that were in its tri-party book on September 12 (Baxter 2010b). The collateral breakdown is shown in Figure 3.

Figure 3: LBI Borrowing, September 15–17, 2008

Loan Date	Source of Funding	Type of Funding	Amount (Billions of Dollars)	Share of Collateral Pledged (Percent)							
				UST/ Agency Securities	Agency MBS	Private-Label MBS	Corporate Bonds	Municipal Bonds	ABS	Equity	Other ^a
09/15/2008 ^b	Fed	PDCF	28.0	13.1	7.0	5.0	41.8	10.0	12.2	7.6	3.3
09/16/2008 ^c	Fed	PDCF	19.7	6.5	0.0	9.6	53.9	1.7	14.9	9.0	4.4
09/17/2008 ^d	Fed	PDCF	20.4	16.4	13.3	2.1	31.5	0.4	16.6	18.6	1.1
09/15/2008	Barclays	Tri-party repo	15.8				Not known				
09/16/2008	Barclays	Tri-party repo	15.8				Not known				
09/17/2008	Barclays	Tri-party repo	15.8				Not known				

Source: Fleming and Sarkar 2014.

The FRBNY also imposed steeper haircuts on LBI's collateral because of the bankrupt status of its parent. Baxter said that the increased haircuts accounted for LBI's "diminished creditworthiness resulting from the loss of its parent's support" (Baxter 2010b, 2). The FRBNY imposed on LBI a haircut of 20% on all non-US-government collateral, while other borrowers saw lower haircuts for highly rated private-label securities (Burke 2008; Fed 2020). Haircuts on LBI's Treasury and agency collateral, except for agency real estate mortgage investment conduits and collateralized mortgage obligations, did not exceed 8% (Burke 2008).

William Dudley stated that the 20% haircut on non-Treasury and non-agency collateral was approximately three times higher than the market was offering, an amount that gave the FRBNY a "fair degree of protection" (Fed 2008a, 7). Moreover, that the PDCF loan was helping to bridge LBI to being purchased by Barclays offered the Fed additional protection, as the Fed would be facing Barclays after the acquisition (Bernanke 2015; Geithner and Metrick 2018; Valukas 2010b).

11. Impact on Monetary Policy Transmission: To control the federal funds rate, the Fed sterilized PDCF loans by selling securities, and the Treasury announced the Supplementary Financing Program on September 17, 2008.

Between September 17, 2008, and September 24, 2008, the Federal Reserve sold \$71 billion of its traditional securities holdings (FRBC n.d.). Chairman Bernanke details the importance of this process in relation to maintaining the federal funds rate during instances of emergency lending:

The sales of Treasuries drained reserves from the banking system, offsetting the increase in reserves created by our lending. This procedure, known as sterilization, allowed us to make loans as needed while keeping short-term interest rates where we wanted them. (Bernanke 2015, 237)

The Treasury's Supplementary Financing Program, announced on September 17, 2008, was also intended to help control the federal funds rate. Treasury issued short-term bills in excess of its regular borrowing program, depositing the proceeds at the Federal Reserve (Fed 2009d).

Even with these efforts, the increase in use of Fed emergency facilities outpaced its ability to drain reserves, weakening the Fed's control over the federal funds rate (Kohn and Sack 2020). Between mid-September and December 2008, when the Fed lowered rates to the zero lower bound and recalibrated its new policy of paying interest on reserves, the effective funds rate was consistently 50 basis points below the target rate (Fed 2024; Kohn and Sack 2020).

12. Other Conditions: The PDCF required borrowers to provide Fed authorities with regular information concerning their financial condition.

The PDCF required borrowing firms to provide the Fed with regular information about their financial condition. This provided Fed authorities with visibility into non-depository firms' balance sheets that had not been previously available (Alvarez, Dudley, and Liang 2020).

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